

Cover and stego images across the imperceptibility and robustness frontier

Cover QR
(no hidden data)



Imperceptible
(clean-trained)
PSNR 54.9 dB, SSIM 1.000, public OK



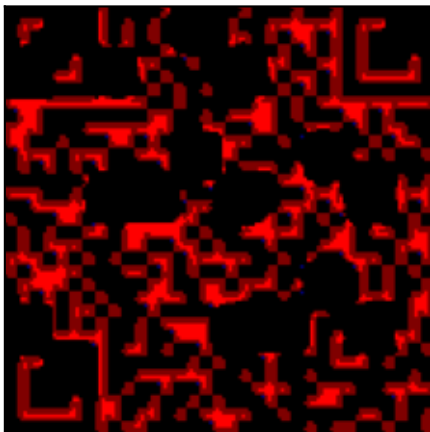
Robust
(distortion-trained)
PSNR 22.5 dB, SSIM 0.988, public OK



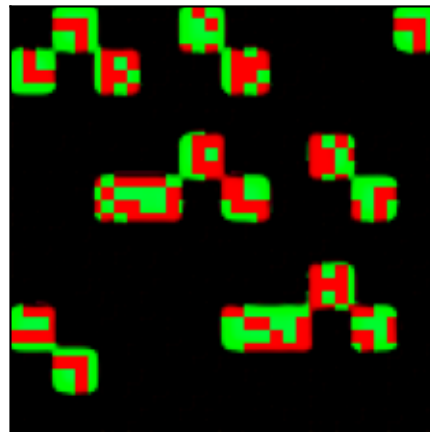
Hybrid robust
(structural mask)
PSNR 20.4 dB, SSIM 0.980, public OK



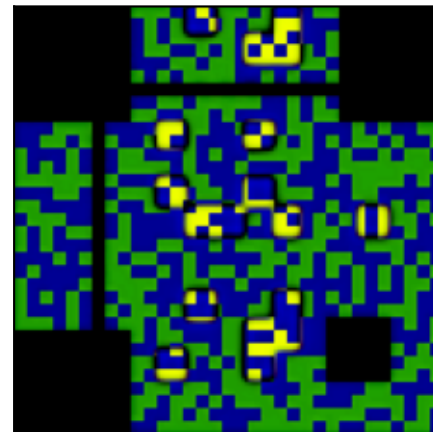
normalized residual, max change 0.01



normalized residual, max change 0.30



normalized residual, max change 0.30



top: stego QR
(public still scans)

bottom: absolute residual
normalized to show
where bits are embedded